Day 4. Gradient descent

Gradient decsent is an algorithm that you can use to find the best parameters for the lost function.

Gradient descent is the algorithm that you can use to minimize any cost functions more than 2 parameters.

Gradient descent

Outline

Start with some w, b
keep changing w, b to reduce J(w, b)
Until we settle at or near a minimum

> recall the gradient descent we have just learn recently.

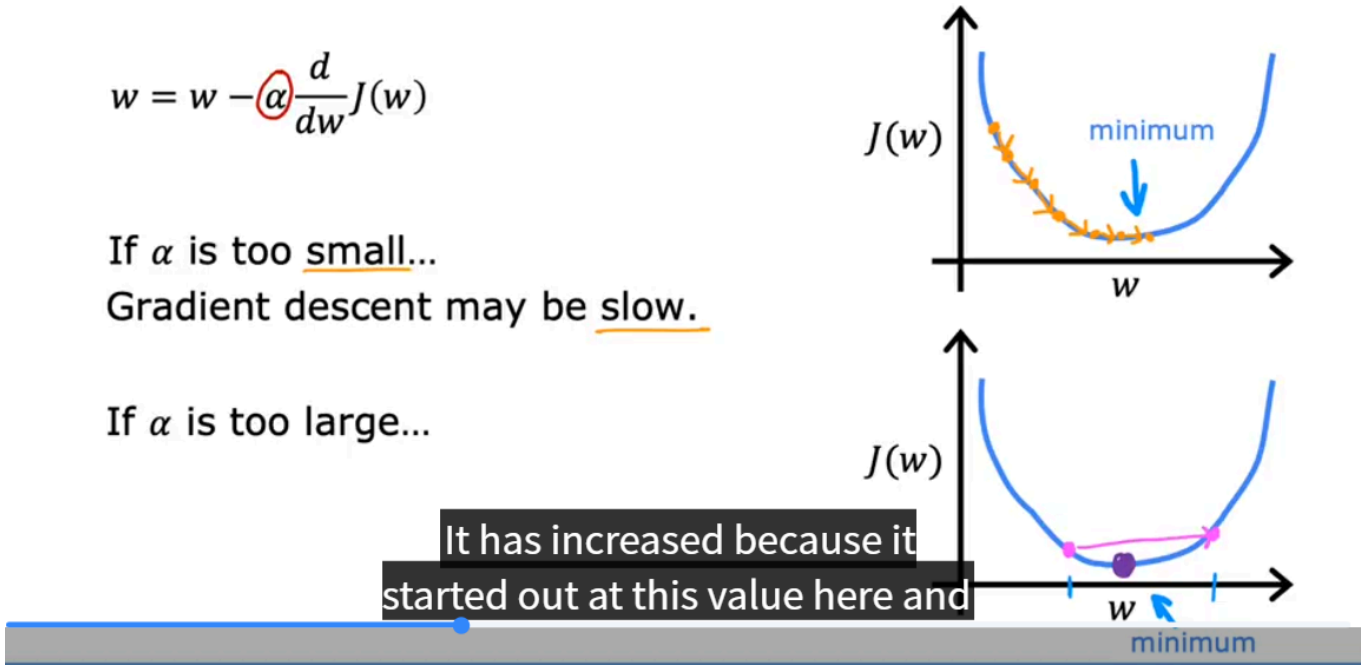
Implementing gradient descent algorithm

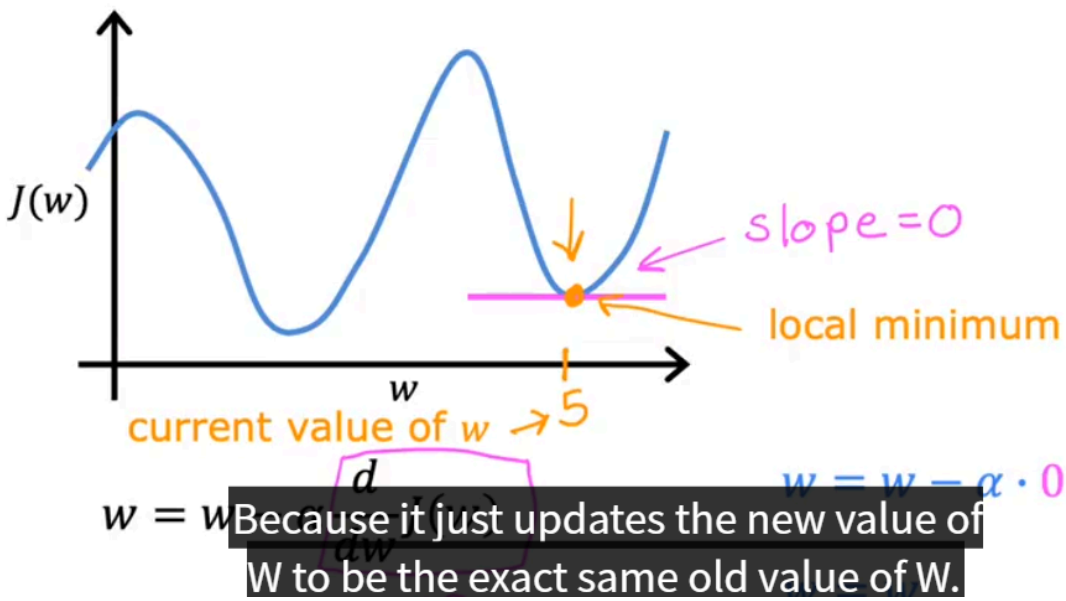
$$w = w - \alpha \frac{\partial}{\partial w} J(w, b)$$

$$b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$

- α: learning rate, basically the "step" you take every time to go "downhill"

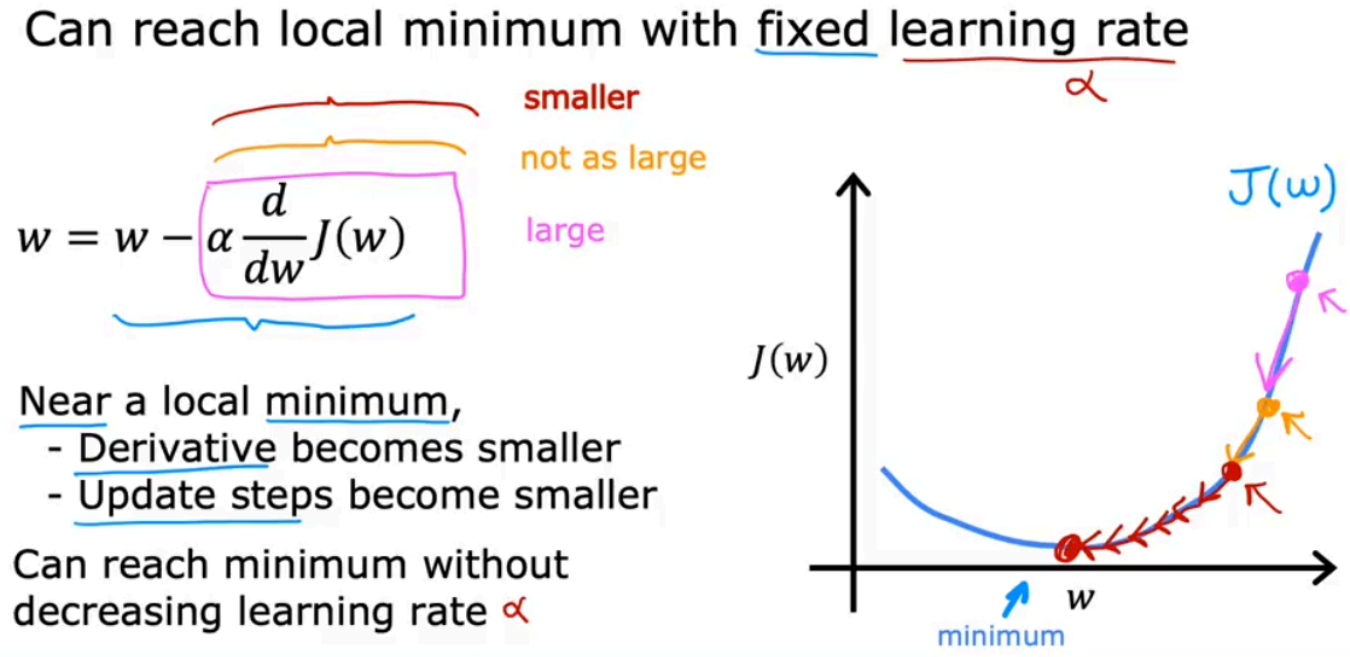
  > note, this is a concurrent process, which means the process goes on the same time.



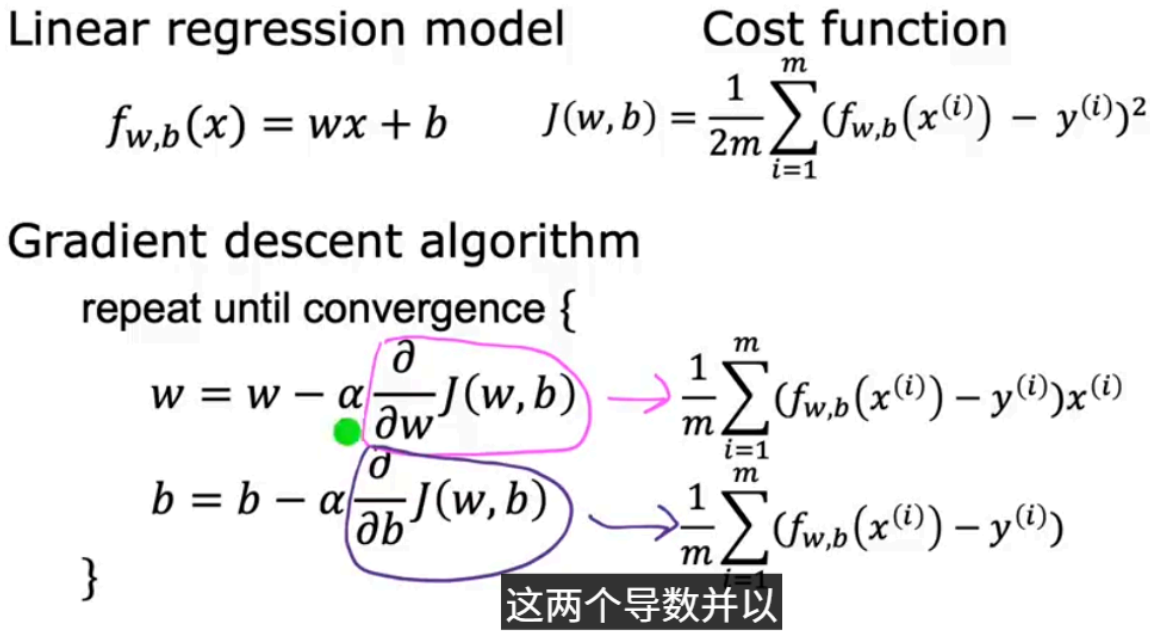


Question

how to implement derirative funtion in programming language?



Gradient descent for linear regression



> well, I start to understand that programming is not the most important thing here, in fact, math intuition is.

"Batch" gradient descent

"Batch": Each step of gradient descent uses all the training examples.

	x size in feet²	y price in \$1000's
(1)	2104	400
(2)	1416	232
(3)	1534	315
(4)	852	178
…	…	…
(47)	3210	870

$$\sum_{i=1}^{m} \left(f_{w,b}\left(x^{(i)}\right) - y^{(i)}\right)^2$$

时，在计算从 i =1 到 m 的总